

## 第29讲. Fourier 级数.

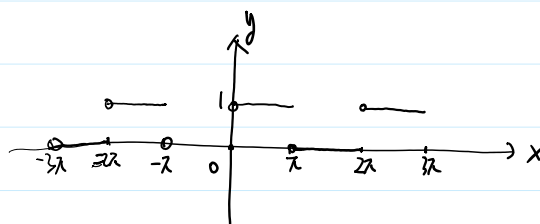
作业: P280. 3. 4.

定义1. 设  $f(x)$  定义在  $[a, b]$  上. 若  $\exists$  分划  $a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$ , s.t. $f(x)$  在  $(x_{k-1}, x_k)$  上可微, 且在分点处单侧极限存在, 且单侧导数存在.则称  $f(x)$  在  $[a, b]$  上分段可微.定理1. 设  $f(x)$  以  $2\pi$  为周期且在  $[-\pi, \pi]$  上分段可微. 则  $f(x)$  的 Fourier 级数在  $\forall x \in \mathbb{R}$  收敛

且收敛到  $\frac{f(x^+) + f(x^-)}{2}$ .  $f(x^+) = f(x+0)$  — 右极限.

(Dirichlet 收敛定理)

$$\text{例1. } f(x) = \begin{cases} 0, & -\pi < x \leq 0 \\ 1, & 0 < x \leq \pi \end{cases}$$

①. 求  $f(x)$  的 Fourier 级数.②. 将  $f(x)$  在  $[-\pi, \pi]$  上展开成 Fourier 级数.

$$\text{解: ①. } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} 1 dx = 1.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} \cos nx dx = 0, \quad \forall n \geq 1.$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{\pi} \sin nx dx = \left. -\frac{\cos nx}{n\pi} \right|_0^{\pi} = \frac{1 - (-1)^n}{n\pi} = \begin{cases} 0 & n=2k \\ \frac{2}{n\pi} & n=2k+1 \end{cases}$$

$$\therefore f(x) \sim \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{(2n-1)\pi} \sin(2n-1)x.$$

②. 由于  $f(x)$  在  $[-\pi, \pi]$  上分段可微.  $\therefore$  由 Dirichlet 收敛定理.

$$\frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{(2n-1)\pi} \sin(2n-1)x = \begin{cases} f(x) & |x| < \pi, x \neq 0, \\ \frac{1}{2} & x = 0, \pm\pi. \end{cases}$$

## 2. 奇、偶函数的 Fourier 级数.

①  $f(x)$  以  $2\pi$  为周期  $\overset{[-\pi, \pi]}{\text{奇函数}}$ .

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx, \quad n \geq 1.$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = 0, \quad n \geq 1.$$

$$\therefore f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{—— 余弦级数.}$$

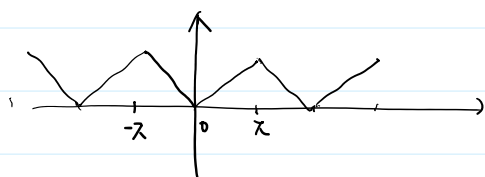
(2)  $f(x)$  以  $2\pi$  为周期且在  $[-\pi, \pi]$  上奇函数.

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = 0 \quad \forall n=0, 1, 2, \dots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx, \quad n=1, 2, \dots$$

$$\therefore f(x) \sim \sum_{n=1}^{\infty} b_n \sin nx \quad \text{—— 正弦级数.}$$

例2. 将  $f(x) = |x|$  在  $[-\pi, \pi]$  上展开成 <sup>以  $2\pi$  为周期</sup> Fourier 级数



解:  $b_n = 0, \quad \forall n=1, 2, \dots$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx = \frac{2}{\pi} \int_0^{\pi} x \, dx = \pi.$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx = \frac{2}{n\pi} x \sin nx \Big|_0^{\pi} - \frac{2}{n\pi} \int_0^{\pi} \sin nx \, dx \\ &= \frac{2}{n^2\pi} \cos nx \Big|_0^{\pi} = \frac{2}{n^2\pi} ((-1)^n - 1) \\ &= \begin{cases} -\frac{4}{n^2\pi} & n=2k-1 \\ 0 & n=2k \end{cases} \end{aligned}$$

$$\therefore f(x) \sim \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)x$$

$\therefore f(x)$  在  $[-\pi, \pi]$  上分段可微的连续函数,  $\therefore$  由 Dirichlet 收敛定理,

$$|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)x, \quad \forall x \in [-\pi, \pi]$$

$$\sim \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)x$$

$$x=0, \quad \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

$$S = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$S_2 = \sum_{n=1}^{\infty} \frac{1}{(2n)^2}$$

$$S_3 = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$$

$$S_1 = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

$$S_2 = \frac{1}{4} S = \frac{1}{4} (S_1 + S_2) \Rightarrow 4S_2 = S_1 + S_2 \Rightarrow S_2 = \frac{\pi^2}{24}$$

$$S = 4S_2 = \frac{\pi^2}{6}.$$

$$S_2 + S_3 = S_1 \Rightarrow S_3 = \frac{\pi^2}{8} - \frac{\pi^2}{24} = \frac{\pi^2}{12}.$$

3. 将  $f(x)$  在  $[0, \pi]$  上展开成 Fourier 级数,

偶式展开

①. 将  $f(x)$  在  $[0, \pi]$  上延拓为  $[-\pi, \pi]$  上偶函数.

$$b_n = 0, \quad n=1, 2, \dots$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx, \quad n=0, 1, 2, \dots$$

$$\therefore f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

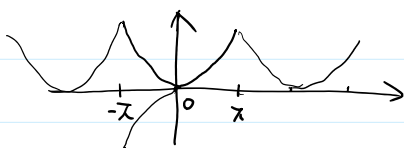
② 奇式展开. 将  $f(x)$  在  $[0, \pi]$  上延拓为  $[-\pi, \pi]$  上奇函数.

$$a_n = 0, \quad n=0, 1, 2, \dots$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx, \quad n=1, 2, \dots$$

$$\therefore f(x) \sim \sum_{n=1}^{\infty} b_n \sin nx$$

例 3. 将  $f(x) = x^2$  在  $[0, \pi]$  上展开成余弦级数.



解:  $b_n = 0, \quad \forall n=1, 2, \dots$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 \, dx = \frac{2}{3} \pi^2.$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2}{3} \pi^2.$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx = \frac{2}{\pi n} x^2 \sin nx \Big|_0^{\pi} - \frac{2}{n^2} \int_0^{\pi} 2 \sin nx \cdot x dx$$

$$= \frac{4}{n^2} x \cos nx \Big|_0^{\pi} - \frac{4}{n^2} \int_0^{\pi} \cos nx dx$$

$$= \frac{4(-1)^n}{n^2}$$

$$\therefore f(x) \sim \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$$

$\therefore f(x)$  在  $[-\pi, \pi]$  上分段可微连续.  $\therefore$  由 Dirichlet 收敛定理.

$$\frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx = x^2, \quad \forall x \in [0, \pi].$$

3. 以  $2l$  为周期的 Fourier 级数.

( $l > 0$ )

设  $f(x)$  以  $2l$  为周期.  $[-l, l]$

$$\text{令 } x = \frac{l}{\pi} y, \quad y = \frac{\pi}{l} x$$

$$f(x) = f\left(\frac{l}{\pi} y\right) = \varphi(y)$$

$$\varphi(y+2\pi) = f\left(\frac{l}{\pi}(y+2\pi)\right) = f\left(\frac{l}{\pi}y + 2l\right) = f\left(\frac{l}{\pi}y\right) = \varphi(y)$$

$$f(x) = \varphi(y) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos ny + b_n \sin ny)$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} \varphi(y) dy = \frac{1}{l} \int_{-l}^l f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \varphi(y) \cos ny dy = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi}{l} x dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \varphi(y) \sin ny dy = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi}{l} x dx$$

$$\therefore f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi}{l} x + b_n \sin \frac{n\pi}{l} x \right)$$

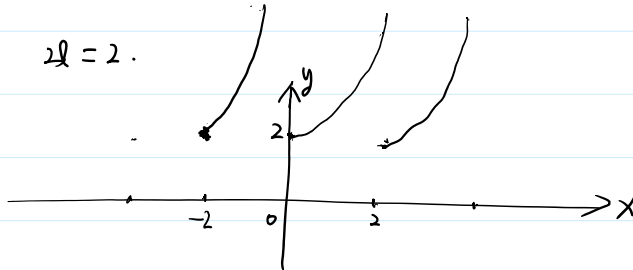
例4. 设  $f(x) = x^2 + x + 2$  在  $[0, 2)$  上的 Fourier 级数展开式

$$S(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x) + b_n \sin(n\pi x)$$

求  $S(0)$ .

解:

$$2l = 2.$$



$$(x + \frac{1}{2})^2 + \frac{7}{4}$$

$$S(0) = \frac{f(0^+) + f(0^-)}{2} = \frac{1}{2} (2 + 8) = 5.$$

= 均方收敛.

$\forall x, y \in \mathbb{R}^n$ .

$$\|x - y\| = \sqrt{\langle x - y, x - y \rangle}$$

$f(x), g(x) \in R[a, b]$ .

$$\|f - g\| = \sqrt{\langle f - g, f - g \rangle}$$

$$s^2(f, g) = \frac{1}{b-a} \int_a^b (f(x) - g(x))^2 dx$$

$\forall x \in \mathbb{R}^n$ .  $\{e_i\} \subset \mathbb{R}^n$  标准基.

$$x = \sum_{i=1}^n x_i e_i$$

$$\|x\|^2 = \sum_{i=1}^n x_i^2$$

设  $f(x)$  以  $2\pi$  为周期. (且  $f(x) \in R[-\pi, \pi]$ ).

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$S_n(f; x) = \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx)$$

$n$  阶部分和.

$$\frac{1}{2} \int_{-\pi}^{\pi} (f(x) - S_n(f; x))^2 dx$$

$$= \frac{1}{2} \langle f(x) - S_n(f; x), f(x) - S_n(f; x) \rangle$$

$$= \frac{1}{2} \langle \underbrace{f(x) - S_n(f; x)}, \underbrace{f(x) - S_n(f; x)} \rangle$$

$$= \frac{1}{2} \left[ \langle \underbrace{f(x) - S_n(f; x)}, f(x) \rangle - \langle \underbrace{f(x) - S_n(f; x)}, S_n(f; x) \rangle \right]$$

$$= \frac{1}{2} \left[ \langle f(x), f(x) \rangle - \underbrace{\langle S_n(f; x), f(x) \rangle}_{\substack{|| \\ 0}} - \underbrace{\langle f(x) - S_n(f; x), S_n(f; x) \rangle}_{|| \\ 0} \right]$$

$$\langle \underbrace{f(x) - S_n(f; x)}, \sin kx \rangle$$

$$= \langle f(x), \sin kx \rangle - \langle \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx), \sin kx \rangle$$

$$= \int_{-\pi}^{\pi} f(x) \sin kx \, dx - \sum_{l=1}^n \left( a_l \underbrace{\int_{-\pi}^{\pi} \cos lx \sin kx \, dx}_{|| \\ 0} + b_l \int_{-\pi}^{\pi} \sin lx \sin kx \, dx \right)$$

$$= \pi b_k - \pi b_k = 0$$

$$\langle f(x) - S_n(f; x), \cos kx \rangle = 0.$$